

Shiva Teja Beerelli

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EDUCATION

The George Washington University (GWU)

Master of Science in Computer Science, *SEAS Dean's Scholarship Award*

August 2023 – May 2025

Washington, DC

- **Coursework:** Data Structures & Algorithm Design, Cloud Computing, Database Management Systems, Software Engineering, Advanced Software Paradigms, Computer System Architecture, Big Data & Analytics, Data Mining, Machine Learning.

B V Raju Institute of Technology (BVRIT)

Bachelor's in Computer Science and Engineering

August 2019 – May 2023

Hyderabad, India

TECHNICAL SKILLS

Programming Languages: Java, C, Python, JavaScript, SQL, NoSQL, HTML, CSS.

Frameworks: React, Spring Framework, Spring Boot, Hibernate, Django, RESTful APIs (Object-Oriented Design)

Databases: MySQL, PostgreSQL, Oracle database, MongoDB.

Cloud Technologies: Amazon Web Services (EC2, IAM, S3, VPC, ECS, RDS, SSM, CloudWatch, Lambda).

Tools: Git, Postman, Jenkins, Maven, Docker, Kubernetes, JUnit, Hadoop, Apache Kafka, Linux, Visual Studio, Eclipse, IntelliJ.

WORK EXPERIENCE

Software Development Engineer | Mety Technology, VA

March 2025 – July 2025

- Contributed to building a social media platform using **Django, React, AWS**, integrating scalable modules into production.
- Integrated secure file upload and deletion using **S3 signed URLs**, enabling realtime UI updates & efficient media handling.
- Revamped API usage to eliminate redundancies, slashing server load by **60%** and boosting dashboard load speed by **40%**.
- Designed and optimized **MySQL** schema on **AWS RDS** to store user and media data, refining performance and scalability.
- Enhanced API reliability via RESTful design and **boto3**, achieving **99.9%** uptime and reducing failed requests by **50%**.

Software Engineer | Burns Law Library, GWU, DC

September 2023 – May 2025

- Developed a web app using **Spring Boot, PostgreSQL** and **React**, automating lookups and cutting manual effort by **40%**.
- Elevated database efficiency by **30%** and cut query response times by **40%** using **partial and multicolumn indexing** in **MySQL**, resulting in faster data retrieval, reduced latency, and enhanced overall system performance.
- Deployed a secure, scalable web application on **AWS Elastic Beanstalk** and **EC2** integrating a high performance database.
- Optimized fullstack performance with **lazy loading, Redis caching, CloudFront CDN**, reducing page load times by **40%**.
- Strengthened **API security** by enforcing **Role-Based Access Control (RBAC)** reducing unauthorized access risks by **50%**.
- Diagnosed **400+** tickets by troubleshooting system issues and configuring cloud services, ensuring smooth operations.

Software Engineer Intern | Mikhvision Digi Solutions LLP, India

January 2023 – July 2023

- Engineered high-throughput **data pipelines**, boosting efficiency by **35%** and ensuring seamless **microservices** integration.
- Improved software quality with **JUnit** and **CI/CD automation**, reducing bugs by **40%** and accelerating releases by **30%**.
- Configured **Jenkins** pipelines for automated builds & deployments with **Docker & Kubernetes**, improving delivery speed.
- Collaborated with a cross-functional Scrum team in an agile environment to manage SDLC tasks & ensure timely delivery.

Full Stack Developer Intern | Jagruti Technosys Pvt. Ltd., India

January 2022 – October 2022

- Assembled high-availability RESTful APIs using Spring Boot, enhancing data processing and improving system efficiency.
- Increased access control efficiency by **50%** by implementing **OAuth 2.0** and **JWT** authentication in **Spring Security**.
- Accelerated database queries by **45%** with **Redis caching** in **Spring Data Redis**, increasing back-end speed by **35%**.
- Streamlined efficiency by reducing **regression testing** time by **40%** through comprehensive **E2E** test cases using **JUnit**.

PROJECTS

Academic Info & Policy Management System

January 2024 - April 2024

- Architected the full-stack software development of an internal tool using Java Spring Boot, React, and MySQL to centralize management of professors, students, courses, and recording policies, enabling real-time lookups & policy enforcement.
- Launched the solution on AWS EC2 and RDS, enabling secure internal access to academic data and policy management.

Fraud Prediction System

January 2024 - April 2024

- Conceptualized a fraud detection system for loan applications using **Big Data Analytics** and **Machine Learning**, deploying a scalable **Hadoop** cluster on **Google Dataproc** with **HDFS** and utilizing parallel processing for efficient model training.
- Implemented data processing and feature engineering with **PySpark**, achieving **99.25%** accuracy and F1-score of **95.23%**.

CERTIFICATIONS & LEADERSHIP ACTIVITIES

Certifications: AWS Cloud Foundations (AWS) | Core Java (Internshala) | Java Spring with Spring Boot (Udemy) | Web Development (Coursera) | Python Programming (Coursera) | Operating Systems (Google Coursera).

Leadership Activities: Cultural Club Head, BVRIT | Coding Brigade BVRIT Club | Hostel Student Representative, BVRIT.